

## Assignment : SMT Encoding

Software Verification CS6410

Each question has 5 marks

Note you are allowed to discuss and take online help.

### Q1. Circuit Equivalence Checking using Uninterpreted Functions (UF)

**Problem:** Given two boolean circuits represented by logic gates, we want to check if they are functionally equivalent for all possible inputs. Check the Equivalence of Two Circuits thus proving the correctness of De Morgan's Law

#### 1. Circuit 1:

- $f_1(A, B) = \text{NOT } (A \text{ AND } B)$

#### 2. Circuit 2:

- $f_2(A, B) = (\text{NOT } A) \text{ OR } (\text{NOT } B)$

These should be **equivalent** since De Morgan's law states:

$$\neg(A \wedge B) = \neg A \vee \neg B$$

Encode the above two circuits in SMTLib and check the correctness of the above formulas using Z3/CVC5

### Q2. Cake cutting at a party

**Problem:** A cake is to be divided among  $n$  players such that no player prefers another player's portion over their own. This is called an envy-free division.

Mathematical Model:

Each player  $P_i$  has a valuation function  $V_i(x)$  that assigns a value to any piece  $x$ .

The goal is to divide the cake into  $n$  disjoint portions  $X_1, X_2, \dots, X_n$  so that:

1. Each player values their own piece at least as much as any other player's piece:  $V_i(X_i) \geq V_i(X_j) \quad \forall i, j$
2. The sum of the pieces equals the whole cake:  $X_1 + X_2 + \dots + X_n = 1$

### Q3. Job Scheduling

#### Problem:

A factory needs to schedule five jobs J1,J2,J3,J4,J5 within a 24-hour work window. The jobs have the following durations:

- J1 = 4 hours
- J2 = 6 hours
- J3 = 5 hours
- J4 = 3 hours
- J5 = 2 hours

#### Constraints:

1. Precedence Constraints:
  - J1 must start before J2.
  - J3 must start after J2.
  - J5 must start after J4.
2. Non-Overlap Constraints:
  - No two jobs can overlap.
  - The total working time cannot exceed 24 hours.
3. Break Time Constraint:
  - A mandatory 1-hour break must be scheduled between job transitions (if a job ends at time  $t$ , the next job can only start at  $t+1$  at the earliest).

Encode this scheduling problem in SMT-LIB and check if a feasible schedule exists.

### Q4. SMTLib over theory of Strings: Password Validation with Constraints

**Problem:** A system requires a password that meets the following criteria:

- At least **8 characters long**
- Contains **at least one digit (0-9)**
- Contains **at least one uppercase letter (A-Z)**
- Contains **at least one special character (@, #, \$, %)**

Model this as an SMT-LIB problem and check if "A2b@cD" is a valid password.

## Q5: Quantified Formulas: Path in a Grid with Obstacles

**Problem:** We have a **5×5 grid** where a robot starts at **(0,0)** (top-left) and must reach **(4,4)** (bottom-right) while avoiding obstacles. The robot can move:

- **Right (R)** →  $(x+1, y)$
- **Down (D)** →  $(x, y+1)$
- **Diagonally Down-Right (DR)** →  $(x+1, y+1)$

Obstacles block movement through certain cells, e.g.: say (2,2) and (3,3)

Model this as an SMT-LIB problem to find some strategy for the problem.

## Q6. Program Synthesis as SMT: Integer Division with Remainder

**Problem:**

We want to **synthesize two functions**,  $q(x,y)$  and  $r(x,y)$ ,

- that compute **integer division and remainder** when dividing  $x$  by  $y$  (i.e., implementing the division algorithm without using `/` or `%`).

For any integers  $x$  and  $y$  (where  $y \neq 0$ ), the functions must satisfy:

$$x = q(x, y) \times y + r(x, y)$$

$$0 \leq r(x, y) < |y|$$

This means:

- $q(x, y)$  should return the quotient  $x \div y$
- $r(x, y)$  should return the remainder  $x \bmod y$
- The remainder  $r(x, y)$  should always be **non-negative** and **less than**  $|y|$

Encode this problem as an **SMT solving problem** (Hint use theory of **Uninterpreted Functions**)